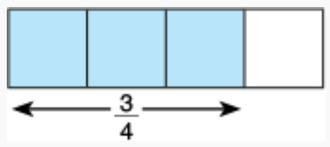
Multiplying Fractions

When you multiply a fraction by a fraction, you are taking something small and breaking it into even smaller parts. You can expect the answer to be smaller than either of the fractions you started with.

Example 1

Hide

Sylvia had $\frac{3}{4}$ pound of candy. She wanted to give each of her 3 grandchildren $\frac{1}{3}$ of the candy. How much candy should Sylvia give each grandchild?



Step 1

Multiply $\frac{3}{4}$ by $\frac{1}{3}$. Look for ways to cancel (or factor out) a multiple common to the numerator and the denominator. This makes the problem simpler.

$\frac{3}{4} \times \frac{1}{3} \longrightarrow \frac{3}{4} \times \frac{1}{3} = \frac{1}{4}$

Step 2

Cancel the numerator of the first fraction and the denominator of the second fraction.

$\frac{1\not{3}}{4} \times \frac{1}{\not{3}_1}$

Step 3

Multiply across using the new numerators and denominators.

$\frac{1}{4} \times \frac{1}{1} = \frac{1}{4}$ pound

Example 2

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Multiply $\frac{6}{7}$ by $\frac{7}{9}$.

Step 1

Look for ways to cancel. Notice that canceling may include both numerators and denominators.

$\frac{6}{7} \times \frac{7}{9} \longrightarrow \frac{{}^{2}\not{6}}{{}_{1}\not{7}} \times \frac{\not{7}^{1}}{\not{9}^{3}} = \frac{2}{1} \times \frac{1}{3} = \frac{2}{3}$

Step 2

Cancel the numerator of the first fraction and the denominator of the second fraction.

$\frac{{}^{2}\not{6}}{7} \times \frac{7}{\not{9}_{3}}$

Step 3

Cancel the numerator of the second fraction and the denominator of the first fraction.

$\frac{{}^{2}\not{6}}{{}_{1}\not{7}} \times \frac{\not{7}^{1}}{\not{9}_{3}}$

Step 4

Multiply across using the new numerators and denominators.

$\frac{2}{1} \times \frac{1}{3} = \frac{2}{3}$

If you did not cancel the fractions before multiplying, you would still get the same answer. However, the answer would have a larger numerator and denominator, and you would have to simplify your answer after multiplying. Therefore, it is much easier to cancel first.

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Lesson 5: Operations with Fractions

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